

Hospital Management System Project SW332-Part2

Part 2 of 3: First Sprint

1. Sprint Planning Report

Sprint Planning Report

Sprint: phase 2 - sprint 1 **Date:** October 16 **Product:** Hospital Management System

1. Sprint Goal:

The team aims to build the core functionalities of the Hospital Management System to ensure smooth workflow between doctors, patients, and administrators — including account registration, appointment booking, and access to patient medical records.

2. Selected Product Backlog Items (with Acceptance Criteria):

- **User Story 1:**

- **Title:** As an Administrator, I want to register hospital staff so that I can manage user access and system roles.
- **Description:** Build a module to register doctors, nurses, and staff in the system.
- **Acceptance Criteria:**
 - Admin can add, edit, and delete hospital staff accounts.
 - The system assigns appropriate roles and permissions automatically.
 - Registered staff can log in successfully.

- **User Story 2:**

- **Title:** As a Doctor, I want to view full medical records of patients before treatment so that I can make accurate diagnoses and decisions.
- **Description:** Build a secure interface for doctors to access patients' full medical history.
- **Acceptance Criteria:**
 - Doctor can view patient information, test results, and diagnosis history.
 - Access is restricted to authorized doctors only.
 - Data is displayed clearly and retrieved without delay.

- **User Story 3:**
 - **Title:** As a Patient, I want to register and create an account so that I can book appointments and access my medical records.
 - **Description:** Build an online registration and login interface for patients.
 - **Acceptance Criteria:**
 - Patient can register successfully with valid information.
 - System validates input (email, ID, password).
 - Patient can access dashboard and view personal information.
- **User Story 4:**
 - **Title:** As a Doctor, I want to view my patient list and appointments so that I can manage my daily schedule efficiently.
 - **Description:** Create an interface to view daily appointments and assigned patients.
 - **Acceptance Criteria:**
 - Doctor can view all appointments by date/time.
 - Schedule updates automatically when patients book or cancel.
 - Doctor can filter appointments by patient or date.
- **User Story 5:**
 - **Title:** As a Patient, I want to book an appointment online so that I can choose a convenient time with my preferred doctor.
 - **Description:** Implement online booking and doctor availability check.
 - **Acceptance Criteria:**
 - Patient can view available slots per doctor.
 - Appointment booking confirms successfully.
 - Notifications are sent to both patient and doctor.

3. Sprint Backlog Breakdown (Tasks):

- **User Story 1 - Administrator Registration:**

Task	Assigned	Estimated Effort
Build interface for registering hospital staff	Roaa	3
Integrate the system with database	Bushra	3

- **User Story 2 - Doctor Views Patient Records:**

Task	Assigned	Estimated Effort
Design interface for displaying patient records	Joud	3
Develop database connection for records retrieval	Bushra	2

3. Sprint Backlog Breakdown (Tasks):

- **User Story 3 - Patient Account Registration**

Task	Assigned	Estimated Effort
Develop registration functionality	Joud	3
Build database integration retrieval	Bushra	4
Create interface to display patient dashboard	Sumaiya	3

- **User Story 4 – Doctor Appointment List**

Task	Assigned To	Estimated Effort
Develop UI for viewing patients and appointments	Sumaiya	4
Implement search/filter by date or patient	Bushra	3

3. Sprint Backlog Breakdown (Tasks):

- **User Story 5 – Patient Appointment Booking**

Task	Assigned To	Estimated Effort
Design UI for selecting doctor and time	Roaa	4
Develop backend API for doctor availability	Sumaiya	2

Total Estimated Effort: 61 Hours **Hours Used:** 43 **Hours Remaining:** 18

- **4. Sprint Capacity and Velocity:**

- **Team Capacity:** 61 hours (≈ one week for 4 members)
- **Estimated Velocity:** 5 user stories per sprint
- **Any Factors Affecting Capacity:** No holidays, but tasks differ by complexity (UI tasks require testing time).
- **5. Risks and Impediments:**
-

Risk	Impact	Mitigation
Integration delay between modules	High	Early API testing and internal review
Database connection errors	Medium	Backup database and error logs
Miscommunication in daily meetings	Low	Use Teams daily and assign Scrum. Aster follow-ups

- **6. Definition of "Done":**
- All acceptance criteria are met.
- Code is tested and integrated.
- UI is functional and connected to backend.
- Tasks are marked Complete in Sprint Backlog..
- System demo is ready for Sprint Review.

3. Sprint Backlog.

SPRINT BACKLOG

CURRENT DATE	SPRINT START DATE	TOTAL ASSIGNED HOURS	HOURS USED	HOURS REMAINING
11/04/25	10/31/25	47	28	19

SPRINT GOAL

Write A short statement of what the team aims to achieve during the sprint.

SPRINT TRACKING

User Stories and Tasks	ASSIGNED TO	ASSIGNED HOURS	HOURS USED	AVAILABLE HOURS REMAINING	Start of the Sprint Date													
					DAY 1	DAY 2	DAY 3	DAY 4	DAY 5	DAY 6	DAY 7	DAY 8	DAY 9	DAY 10	DAY 11	DAY 12	DAY 13	DAY 14
User Story 1 Name: As an accountant, I want to generate billing and payment reports so that I can track hospital revenue and patient		0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Task 1.1 Create Billing page	Member's Name: Sumaiya	5	3	2	2	2	1	0	0	0	0	0	0	0	0	0	0	0
Task 1.2 Link billing to database	Member's Name: Bushra	3	2	1	0	1	1	1	0	0	0	0	0	0	0	0	0	0
User Story 2 Name: As a doctor, I want to request lab tests and view results digitally so that I can make faster medical decisions.		0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Task 2.1 Design lab test request form	Member's Name: Bushra	3	2	1	2	1	0	0	0	0	0	0	0	0	0	0	0	0
Task 2.2 Create result display page	Member's Name: Joud	3	2	1	0	1	1	1	0	0	0	0	0	0	0	0	0	0
User Story 3 Name: As a pharmacist, I want to manage and update available medicines in the inventory so that I can ensure		0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Task 3.1 Build Medsout table	Member's Name: Sumaiya	3	2	1	2	1	0	0	0	0	0	0	0	0	0	0	0	0
Task 3.2 Add edit and delete options	Member's Name: Ploaa	3	2	1	0	1	1	1	0	0	0	0	0	0	0	0	0	0
User Story 4 Name: As an administrator, I want to generate reports on hospital performance and staff activities so that I can evaluate efficiency and		0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Task 4.1 Create admin performance dashboard	Member's Name: Joud	3	2	1	2	1	0	0	0	0	0	0	0	0	0	0	0	0
Task 4.2 Add staff activity data	Member's Name: Ploaa	3	2	1	0	1	1	1	0	0	0	0	0	0	0	0	0	0
User Story 5 Name: As a patient, I want to pay medical bills online securely so that I can complete my payments easily from home.		0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Task 5.1 Design online payment form	Member's Name: Sumaiya	3	2	1	2	1	0	0	0	0	0	0	0	0	0	0	0	0
Task 5.2 connect payment to database	Member's Name: Joud	2	1	1	0	1	1	0	0	0	0	0	0	0	0	0	0	0
User Story 6 Name: As a patient, I want to view prescriptions and lab results online so that I can stay informed about my health status.		0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Task 6.1 Create lab result page	Member's Name: Ploaa	4	2	2	2	1	1	0	0	0	0	0	0	0	0	0	0	0
Task 6.2 Display prescriptions for patients	Member's Name: Bushra	2	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0
User Story 7 Name: As a patient, I want to receive appointment reminders via SMS or email so that I won't forget my scheduled visits.		0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Task 7.1 Add reminder notification system	Member's Name: Joud	4	2	2	2	1	1	0	0	0	0	0	0	0	0	0	0	0
Task 7.2 Connect reminder to appointment data	Member's Name: Sumaiya	1	0	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0
User Story 8 Name: As a doctor, I want to communicate with other doctors through secure messaging so that I can discuss patient cases		0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Task 8.1 Create simple chat page	Member's Name: Ploaa	3	2	1	2	1	0	0	0	0	0	0	0	0	0	0	0	0
Task 8.2 Enable sending and receiving messages	Member's Name: Bushra	2	1	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL		47	28	19														

4. Sprint Review Report

Sprint Review Report

Sprint: Phase 2 - Sprint 1

Date: November 5

Product: Hospital Management System

Sprint Goal: Build core functionalities to ensure a smooth workflow between doctors, patients, and administrators, including account registration, appointment booking, and access to patient medical records.

I. Sprint Goal Achievement and Completed Work:

1.1 Sprint Goal Review:

- Sprint Goal Achieved: Yes

- Achievement Details: The team successfully developed the core functionalities required for the system, including hospital staff registration, patient account creation, appointment booking, and doctor access to medical records. The modules are fully integrated and working seamlessly.

1.2 Completed Work Demonstration:

- The Development Team demonstrated the following completed Product Backlog items:

- User Story 1: Administrator registration ○ User Story 2: Doctor views patient records ○ User Story 3: Patient account registration ○ User Story 4: Doctor appointment list ○ User Story 5: Patient appointment booking • Key functionalities and features demonstrated: ○ Administrator can register, edit, and delete staff. ○ Doctors can securely access patient medical records. ○ Patients can register accounts and book appointments online. ○ Doctors can view daily appointments and manage schedules.

- The system confirms appointment bookings and notifies both patients and doctors.

- Any questions from stakeholders were addressed during the demonstration: ○ Stakeholders inquired about the system's security and data privacy measures, which were clarified.

1.3 Acceptance Criteria Verification: •

Verification of Acceptance Criteria: ○

All acceptance criteria were met for each demonstrated item. ○ No deviations or issues were encountered with the acceptance criteria.

II. Product Increment and Stakeholder Feedback:

2.1 Product Increment Discussion:

- **Current State of Product Increment:** The product increment includes all essential features for the hospital management system, with seamless interactions between users (administrators, doctors, patients) and the system.
- **Alignment with Product Goal:** The increment aligns with the overall goal of improving hospital workflow by providing core functionalities that facilitate account management, appointment scheduling, and medical record access.
- **Value to Users/Customers:** This increment significantly improves user experience by automating administrative tasks, providing secure access to medical records, and enabling easy appointment bookings for patients.

2.2 Stakeholder Feedback:

- **Key Feedback Received:**
 - **Usability:** Stakeholders appreciated the clear and user-friendly interface but requested further refinements in the appointment booking system for better flexibility.
 - **Functionality:** Positive feedback was given regarding the accurate display of medical records for doctors, with a suggestion to enhance data retrieval speed.
 - **Business Value:** Stakeholders emphasized the need for more detailed reporting features for administrators to monitor system usage and staff performance.
- **Next Steps Based on Feedback:**
 - Enhance the appointment booking flexibility.
 - Improve the data retrieval speed for patient records.
 - Add reporting features for administrators to track staff performance.

2.3 Product Backlog Review:

- **Product Backlog Adjustments:**
 - Adjustments were made to include improvements based on stakeholder feedback, especially concerning the appointment booking flexibility and reporting features.
- **Newly Discovered Work:**

- Added tasks for enhancing appointment scheduling and creating administrative reports.
- III. Progress and Next Sprint Planning:

3.1 Progress Toward Product Goal:

- Contribution to Product Goal: This sprint contributed significantly to the product goal by building the foundational features of the system, enabling key functionalities like account management and appointment booking.
- Impediments: No major impediments encountered, though some time was spent resolving minor integration issues between modules.

3.2 Next Sprint Planning Preview:

- Potential Items for Next Sprint:
 - Enhancing appointment booking flexibility.
 - Improving patient record retrieval speed.
 - Adding reporting features for administrators.

3.3 Action Items:

- Action Items:
 - Product Owner: Update the Product Backlog with the enhancements for appointment booking and reporting features by October 18.
 - Development Team: Begin implementation of the new features for the next sprint by November 16.

IV. Summary and Closing:

4.1 Summary of Key Takeaways:

- Key Takeaways: The sprint was successful in achieving the goal of building core functionalities for the hospital management system. The system is operational and aligns with the product vision.
- Team Accomplishments: The team successfully completed all planned user stories, and the product now supports essential tasks for users across different roles.

4.2 Next Steps:

- Thank You: Thank you to all attendees for their participation and valuable feedback.
- Next Sprint Planning: Next sprint planning will be on November 13 at 10:00 AM

5. Sprint Retrospective Report

Date:1/11/2025

Meeting facilitator: Roaa alhumaidi

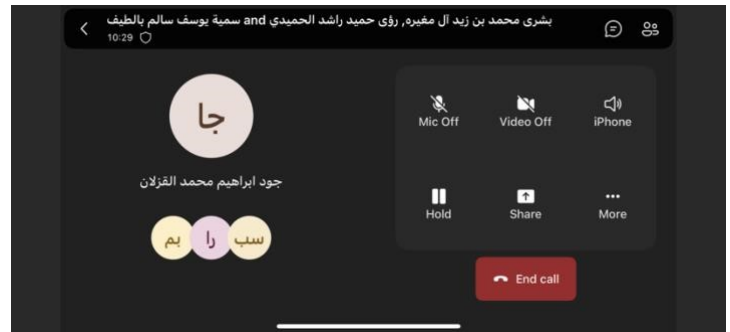
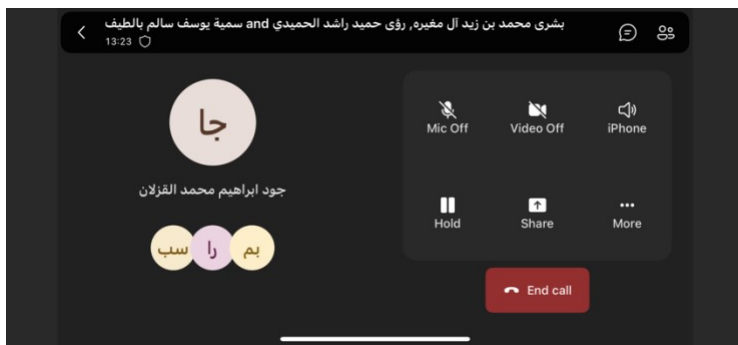
Start	Stop	Continue
1 .End-of-day review: Allocate time at the end of the day to review what has been accomplished and update progress. 2.Test each part of the website: Ensure each part of the website is tested separately after development. 3. Apply INVEST to each User Story: Use the INVEST model for each User Story to ensure clarity of objectives and feasibility.	1 .Non-compliance with daily meetings: Ensure everyone attends daily meetings and is punctual. 2 .Constantly postponing issues: Address issues as soon as they arise, avoiding delays and accumulation of obstacles. 3 .Not reviewing code regularly: Allocate time for regular code reviews to reduce errors and ensure quality work.	1. Continue with daily meetings (Daily Scrum): These meetings help identify each team member's progress and any obstacles they may face. 2.Continue with clear task division: Ensure that each team member knows their assigned tasks and how to execute them effectively. 3 .Continue improving team efficiency: Enhance collaboration and communication among team members to expedite task completion.

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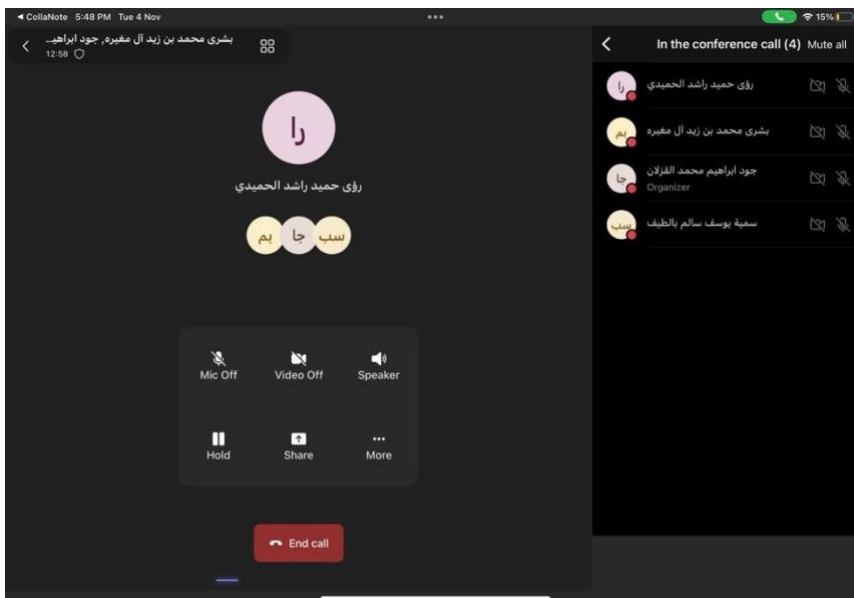
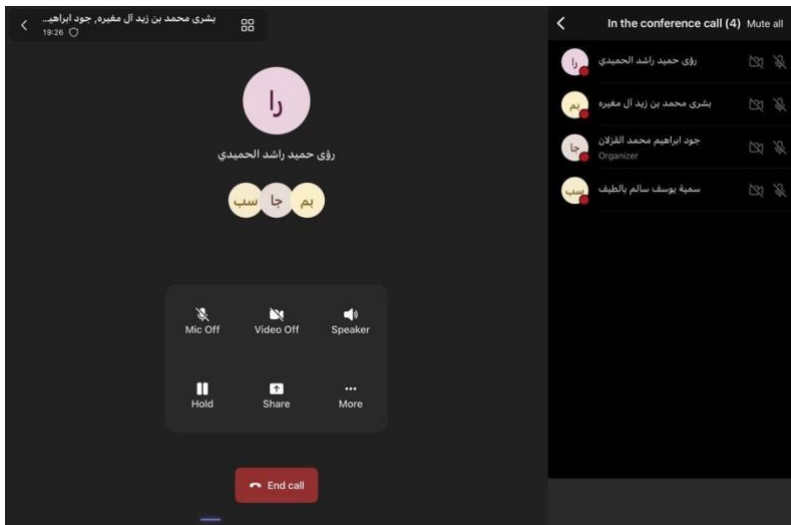
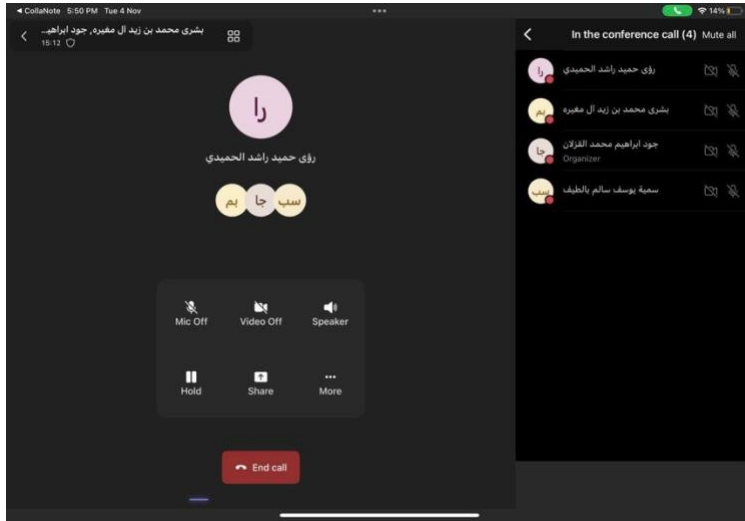
6. Daily Scrum Meetings template.

DAILY SCRUM

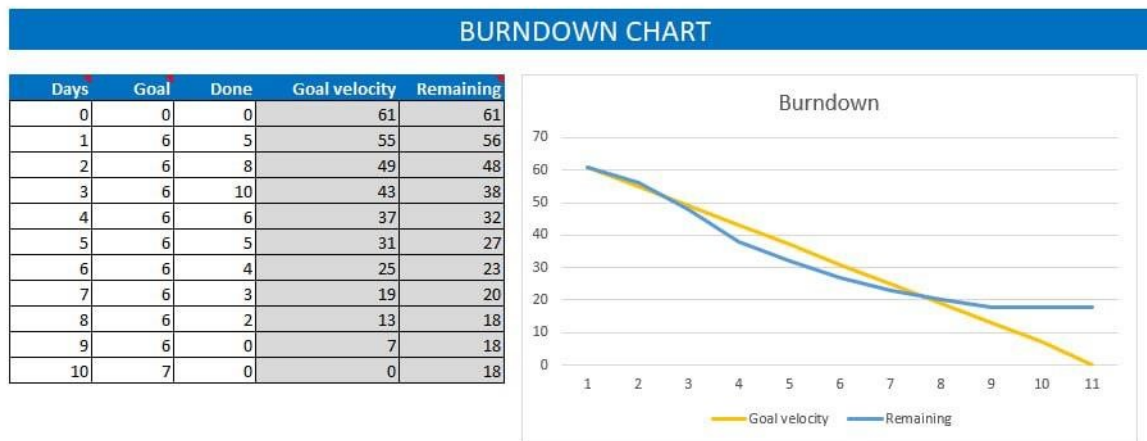
Team member	Question	Sunday	Monday	Tuesday	Wednesday	Thursday
Joud	What did you do yesterday?	coordinated a suitable time for the team members to meet	Start sprint planning meeting	design interface for display patients medical records	develope the functionalities of appointment booking	completed developing the functionalities of appointment booking
	What are doing today?	Start sprint planning meeting	design interface for display patients medical records	develope the functionalitiy of appointment booking	Continue developing the functionalities of appointment booking	Test all my tasks
	Is there anything blocking you?	No	No	Time isn't enough	No	No
Roaa	What did you do yesterday?	improving our work according to the feedback provided by the instructor	Presenting the improvements to the team.	Build interface for registering hospitoal staff	build interface for patients registration	Design UI for selecting doctor and viewing available slots
	What are doing today?	Presenting the improvements to the team.	Build interface for registering hospitoal staff	build interface for patients registration	Design UI for selecting doctor and viewing available slots	Test all my tasks
	Is there anything blocking you?	No	No	No	No	
Sumaiya	What did you do yesterday?	Define what needs to be achieved during the sprint	Complete arrangement of sprint tasks.	design interface to display patients medical record	Develop the functionality to retrieve and display patient list	Complete Developing the functionality to retrieve and display patient list
	What are doing today?	Complete arrangement of sprint tasks.	design interface to display patients medical record	Develop the functionality to retrieve and display patient list	Countinue Developing the functionality to retrieve and display patient list	Test all my tasks
	Is there anything blocking you?	No	No	Network issues	No	No
Bushra	What did you do yesterday?	Update the product backlog	Presenting the new product backlog	integrate the system with database	implement a search /filter for appointment based on date or patient	Develop backend API to fetch doctors' availability
	What are doing today?	Presenting the new product backlog	integrate the system with database	implement a search /filter for appointment based on date or patient	Develop backend API to fetch doctors' availability	Test all my tasks
	Is there anything blocking you?	No	No	No	No	No



SW332 Project



7. Burn Down Chart template.



8. Updated Product Backlog.

SW332 Project

ID	AS A ...	I WANT TO ...	SO THAT ...	PRIORITY	Number of SPRINTS	STATUS
1	Administrator	register hospital staff	I can manage user access and system roles	High	1	Complete
2	Patient	register and create an account	I can book appointments and access my medical records	High	1	Complete
3	Patient	book an appointment online	so that I can choose a convenient time with my preferred doctor	High	2	Complete
4	Doctor	view my patient list and appointments	I manage my daily schedule efficiently	High	1	Complete
5	Doctor	view a patient's full medical record before treatment	so that I can make accurate diagnoses and decisions	High	1	Complete
6	Patient	enter my insurance information in the hospital application	So that I can check if my insurance is valid and see my coverage amount	High	1	In Progress
7	Lab Technician	upload test results to the system	doctors can access lab data without delay	High	2	In Progress
9	Accountant	generate billing and payment reports	I can track hospital revenue and patient payments	Medium	2	Not Started
10	Doctor	request lab tests and view results digitally	I can make faster medical decisions	Medium	2	Not Started
11	Pharmacist	manage and update available medicines in the inventory	I can ensure that prescriptions are filled correctly	Medium	1	Not Started
12	Administrator	generate reports on hospital performance and staff activity	I can evaluate efficiency and improve management	Medium	2	Not Started
13	Patient	pay medical bills online securely	I can complete my payments easily from home	Medium	2	Not Started
14	Patient	view prescriptions and lab results online	I can stay informed about my health status	Medium	2	Not Started
15	Patient	receive appointment reminders via SMS or email	I won't forget my scheduled visits	Low	1	Not Started
16	Doctor	communicate with other doctors through secure messaging	I can discuss patient cases easily	Low	2	Not Started
TOTAL					23	

9. System interfaces:

Hospital Management System

Device

Hospital Management System

HomeLoginAboutContact

Welcome to Our Hospital

A unified platform for hospital management, connecting doctors, patients, and administrators seamlessly.


Login as Admin


Login as Doctor


Login as Patient

Hospital Management System

Create Admin Account

Full Name *

Email *

Password *

Confirm Password *

Create Admin Account

Already have an account? Login

Admin Panel

Dashboard

+ Add Doctor

View Doctors

Manage Appointments

Logout

Administrator Dashboard

Add New Doctor

Full Name

Email

Password
Set a strong password

Department
Cardiology

Contact Number

Add Doctor

Admin Panel

Dashboard

+ Add Doctor

View Doctors

Manage Appointments

Logout

Administrator Dashboard

Doctors List

NAME	DEPARTMENT	EMAIL	ACTION
Dr Norah	Cardiology	norah@gmail.com	
Dr Raaa	Pediatrics	raaa@gmail.com	
Dr Sumaya	Pediatrics	sumaya@gmail.com	
Dr saleh	Radiology	saleh@gmail.com	
Dr john	Radiology	john@gmail.com	

Hospital Management System

[Home](#)[Login](#)[About](#)[Contact](#)

Create Patient Account

Sign up to manage your appointments.

Full Name *

Email *

Password *

Confirm Password *

Date of Birth *

mm/dd/yyyy

Contact Number *

Register

Already have an account? [Login](#)

Patient Portal

Welcome, **haya!** [Logout](#)

[My Appointments](#)

[+ Book Appointment](#)

[Medical Record](#)

Book an Appointment

Select Department

Pediatrics

Select Doctor

Dr Roaa

Select Date

11/21/2025

Select Time

09:30 AM

10:00 AM

11:00 AM

11:30 AM

01:30 PM

02:00 PM

03:00 PM

03:30 PM

Book Appointment

Patient Portal

Welcome, **haya!** [Logout](#)

[My Appointments](#)

[+ Book Appointment](#)

[Medical Record](#)

My Appointments

DOCTOR	DEPARTMENT	DATE	TIME	STATUS	ACTION
Dr Roaa	Pediatrics	2025-11-21	03:00 PM	Upcoming	Cancel

Admin Panel

Dashboard

+ Add Doctor

View Doctors

Manage Appointments

Logout

Administrator Dashboard

Manage All Appointments

PATIENT	DOCTOR	DATE & TIME	STATUS	ACTION
sara	Dr Norah	2025-12-04 at 10:00 AM	Upcoming	
haya	Dr Roaa	2025-11-21 at 03:00 PM	Upcoming	

Doctor Portal

Dashboard

Appointments

Logout

Doctor Dashboard

My Appointments

Search by patient name...

mm/dd/yyyy

Clear Filters

PATIENT	DATE & TIME	STATUS	ACTIONS
haya	2025-11-21 at 03:00 PM	Upcoming	Complete

Doctor Portal

- Dashboard
- Appointments
- Logout

Doctor Dashboard

My Appointments

Search by patient name

PATIENT
haya

Add Medical Record

Patient: haya
Date: 2025-11-21
Diagnosis
Treatment / Procedure
Prescription
Notes (Optional)

Save Record

Patient Portal

Welcome, **haya!** Logout

My Appointments
+ Book Appointment
Medical Record

My Medical Records

التهاب جيوب الأنفية
Date: 2025-11-21
Doctor: Dr Rood

Treatment / Procedure

عمل بخار ماء دافئ مرتين يومياً، واستخدام غسول ملحي للأنف يومياً

Prescription

500mg أموكسيسيلين 500mg كبسولة كل 8 ساعات لمدة 7 أيام باراسيتامول 500mg حبة كل 8 ساعات عند الحاجة لتخفيف الألم

Additional Notes

لمريضه تشعر بالتحسن بسيط بالأنف وصداخ خفيف. يتم مراجعة الحالة بعد أسبوع لتتابعه التحسن